

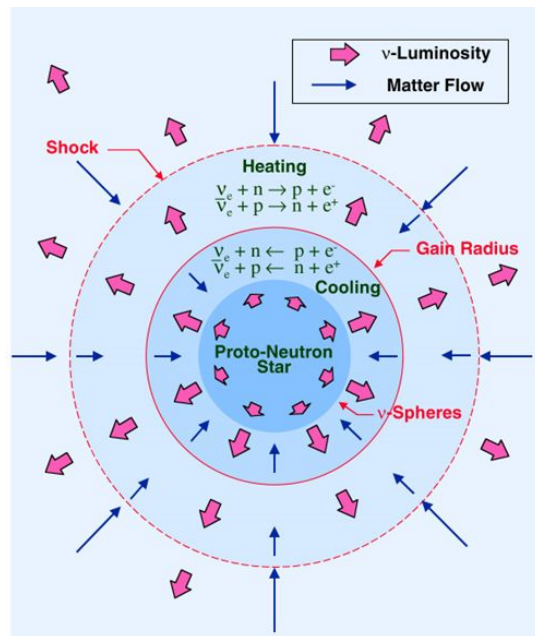
High energy neutrinos from SN explosions

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Overview

- High-energy neutrinos from SNe
- Why is it important to study H-E neutrinos from CCSN?
- Our goal: sensitivity studies for H-E neutrinos from SNe
- First approach: time dependent Unbinned Likelihood using *Aanet*
- Preliminary Studies: model, source catalogue, calculation of expected neutrino events for ARCA21

High-energy neutrinos from SNe



- The core-collapse supernovae (CCSNe) explosion mechanism is driven by low energy (MeV) neutrinos, which release most of the system's gravitational binding energy.
- In addition to MeV neutrinos, some CCSNe are good candidates for the production of high-energy (H-E) neutrinos (from GeV to PeV)

We consider two production mechanisms for H-E neutrinos in CCSNe:

interaction between SN ejecta and CSM
([Murase 2018](#))

relativistic choked jets
([Enberg et al. 2009](#))

Why is it important to study H-E neutrinos from CCSN?

Neutrino telescopes such as the IceCube Neutrino Observatory and KM3NeT observe the burst of MeV neutrinos from CCSNe as single photon hits from interactions that occur near the PMTs.



This measurement is quite sensitive to the high level of single photon noise (dark noise of K40 in the seawater or glass)

In addition, the potential to extend the detection horizon can be done only through detecting H-E neutrinos

H-E ν could give us an insight into the CR acceleration processes (inaccessible with the L-E ν)

SNe CANDIDATES



- CSM interaction candidates: ***SN type IIn and II-P***
- Relativistic CJs candidates: ***SN type Ib/c and IIb***

Sensitivity studies for H-E ν from SNe

OUR GOAL

Estimation of the **sensitivity** for different configurations of ARCA detector to H-E ν coming from SN explosion.

Two approaches to be compared:

1. Time dependent unbinned Likelihood using *Aanet*
2. Cut&Count analysis

We started with preliminary studies that are mandatory for both approaches

- Study the SN neutrinos emission model
- Evaluation of the expected number of neutrino events using the model and different configurations of ARCA

For the moment, we started investigating on the time dependent unbinned analysis...

Time dependent Unbinned Likelihood using AANET

In order to find an excess of ν from given SN positions and times, we chose to follow a time dependent point source likelihood method as described in

[“Constraining High-Energy Neutrino Emission from Supernovae with IceCube”](#)

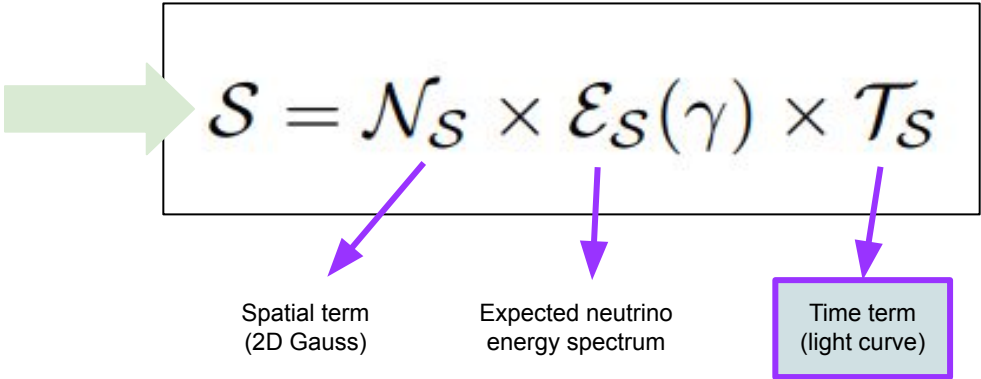
The Likelihood function is given by

$$\mathcal{L} = \prod_{i=1}^N \left(\frac{n_s}{N} \mathcal{S}(\nu_i) + \left(1 - \frac{n_s}{N} \right) \mathcal{B}(\nu_i) \right)$$

Where $\mathcal{S}(\nu)$ and $\mathcal{B}(\nu)$ are PDF respectively for signal and background

Time dependent Unbinned Likelihood using AANET

The signal PDF (same for BKG PDF) is a product of 3 terms


$$\mathcal{S} = \mathcal{N}_S \times \mathcal{E}_S(\gamma) \times \mathcal{T}_S$$

Spatial term
(2D Gauss)

Expected neutrino
energy spectrum

Time term
(light curve)

The signal time PDF $\mathcal{T}_S$ corresponds to the expected neutrino flux as a function of time (light curve); the bkg time PDF $\mathcal{T}_b$ assumes a constant background rate.

Longer neutrino emission would be expected under the scenario of CSM interaction: it would be necessary to test all SN types with box function PDFs of different lengths, up to 1000 days possibly.

The SNe IIn and IIP light curves are going to be tested with the formula:

$$\mathcal{T}(t) \propto (1 + t/t_{\text{pp}})^{-1}$$

Preliminary studies

We started focusing on H-E neutrinos from CSM interaction production mechanism:

1) Model:

The model used for our studies is the one described in the Murase's works ([1], [2])

The neutrino energy fluence is described as follows:

$$E_\nu^2 \phi_\nu \sim \frac{1}{4\pi d^2} \frac{1}{2} \underbrace{\min[1, f_{pp}]}_{\text{pp optical depth}} \underbrace{\frac{\epsilon_{\text{cr}} \mathcal{E}_s}{\mathcal{R}_{\text{cr}10}}}_{\text{Spectrum dependent factor}} \underbrace{\left(\frac{E_\nu}{0.4 \text{ GeV}} \right)^{2-s}}_{\text{Proton momentum index} \text{ and } \text{Spectrum dependent factor}}$$

Fraction of energy carried by CR

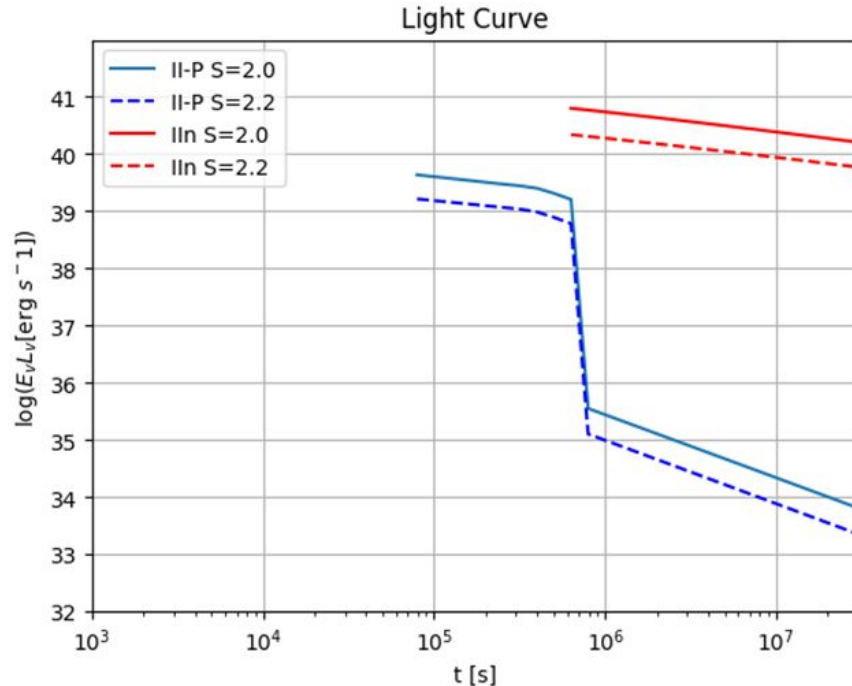
2) Sources catalogue:

Producing the catalogue consists of

- selecting SNe Type II (II_n or II_P);
- choosing the sources according to the life time of ARCA (from ARCA6 to ARCA21);
- selecting the declinations in the range from -90° to 40° (according to KM3NeT field of view)

1.a) Model of the Light Curve

To reproduce the light curves of SN IIn and II-P we used the neutrino flux by Murase's simulation:
(<https://github.com/pegasuskmurase/ModelTemplates/tree/main/SNHEMM>)



Firstly, we managed to reproduce the differential neutrino luminosity vs. time for IIn and II-P type SN
(fig. 1 from the paper [1])

The light curves in the plot are drawn selecting fixed ν energy

$$E_\nu = 1\text{TeV}$$

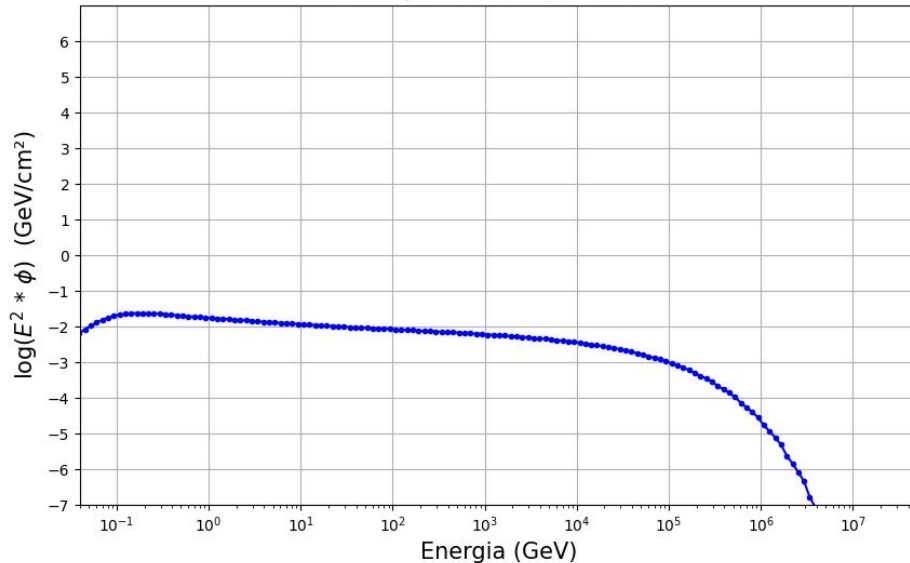
*both proton momentum index (s=2.0 and s=2.2)

1.b) Model of the Fluence

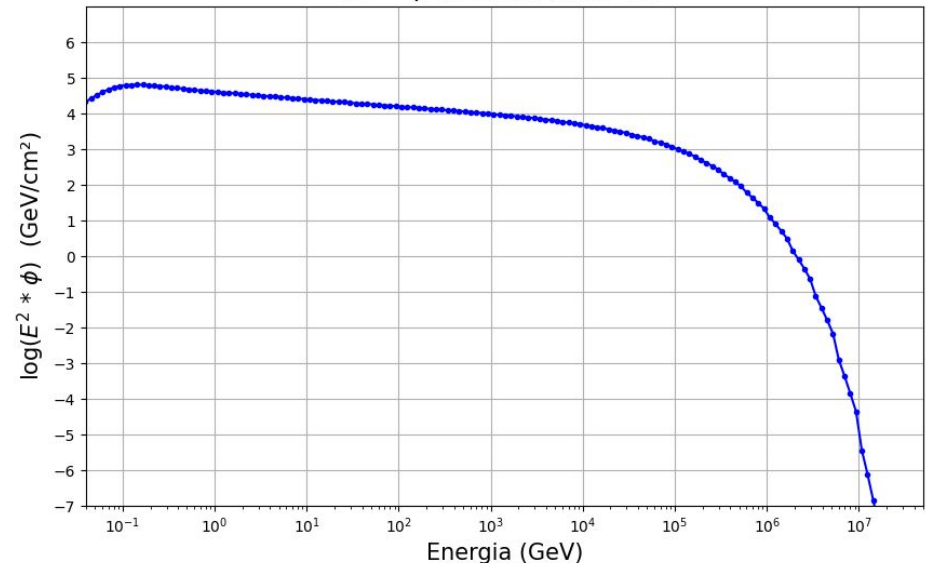
We also reproduced the fluences of neutrinos for both type of SN, considering the time $t_{\nu\text{max}}$ as the time when the significance of neutrino emission is maximized.

(See fig. 2 right from the paper [2])

$E^2 * \phi$ vs E SN II-P k=25



$E^2 * \phi$ vs E SN II n k=26



1.c) Number of expected events

The light curves will be used for the time-dependent unbinned likelihood (signal time PDF)

In the meanwhile, we have been focusing on reproducing the **expected number** of neutrinos, considering time range correspondent to the H-E neutrino emission as predicted in the Murase model.

$$N_\nu = \int_{t_{min}}^{t_{max}} \int_{E_{min}}^{E_{max}} \phi(E_\nu, t) A_{eff}(E_\nu) dt dE_\nu$$

Duration of the H-E
neutrino emission

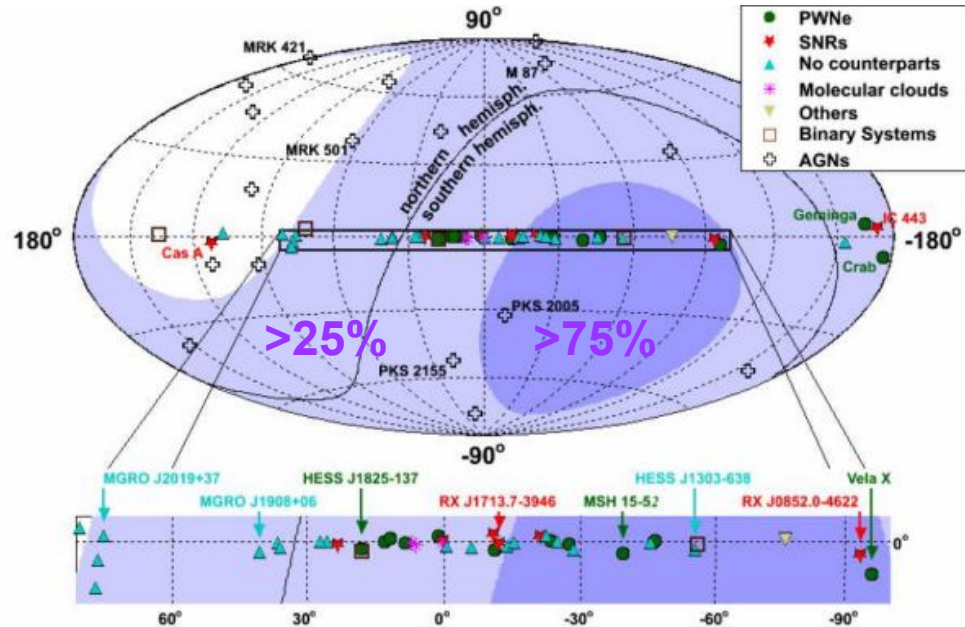
Integrated SN neutrino flux
over time and energy

Integrated effective area
over energy

2.a) ARCA21 effective area for point source

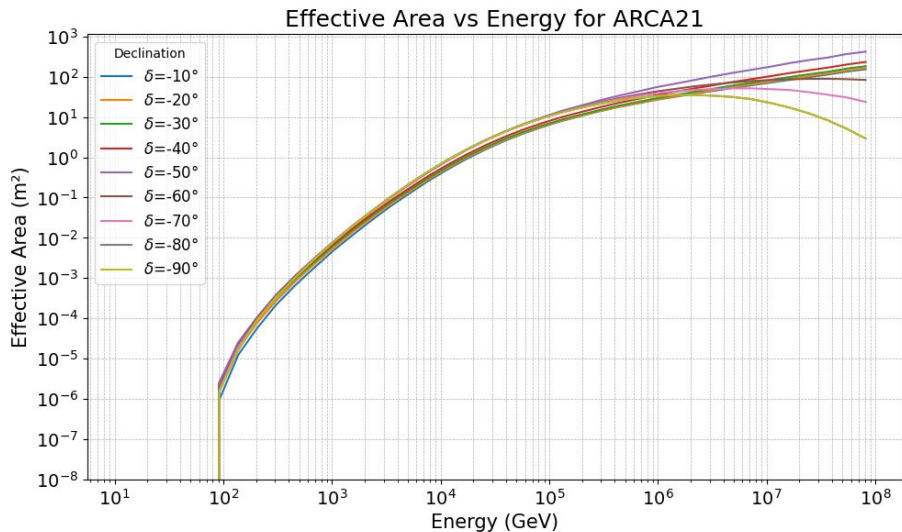
We considered the effective area at different declinations.

In particular, we computed the A_{eff} for the declination in the range $[-90^\circ, 40^\circ]$, considering the southern hemisphere.

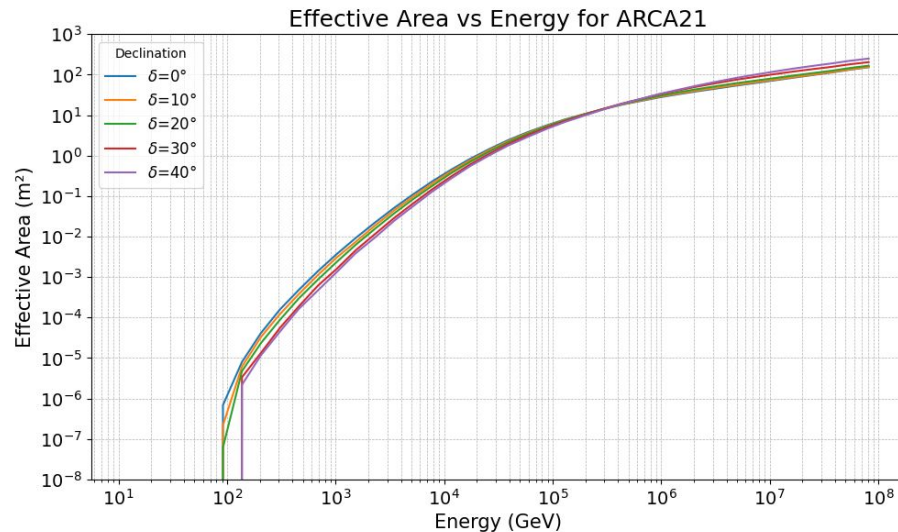


2.a) ARCA21 effective area for point source

Negative declinations

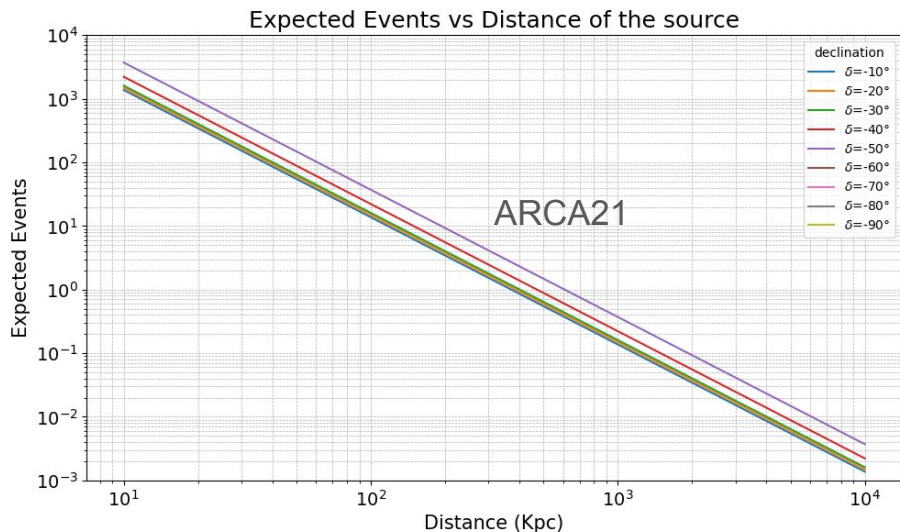


Positive declinations

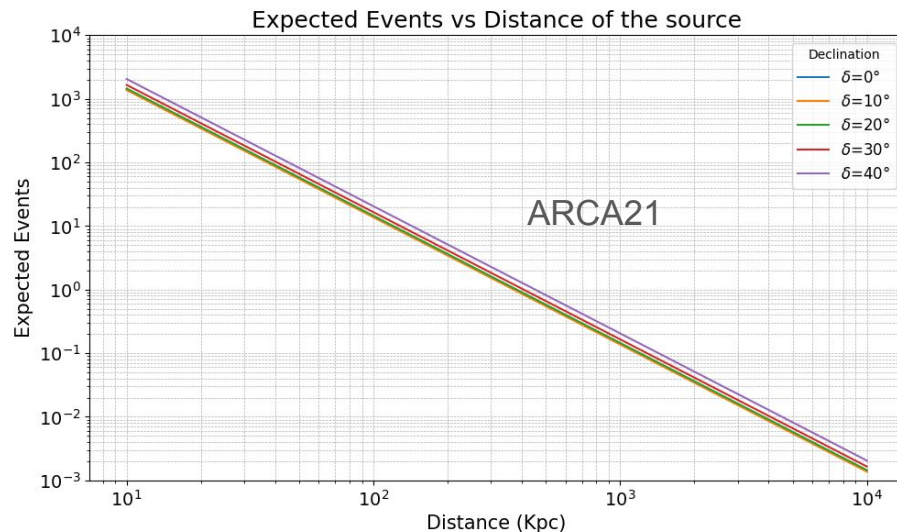


2.b) Number of expected events (SN type II_n)

We chose to plot the number of events vs. the source distance in a range from 10 Kpc to 10 Mpc, for each declination → **SN type II_n**



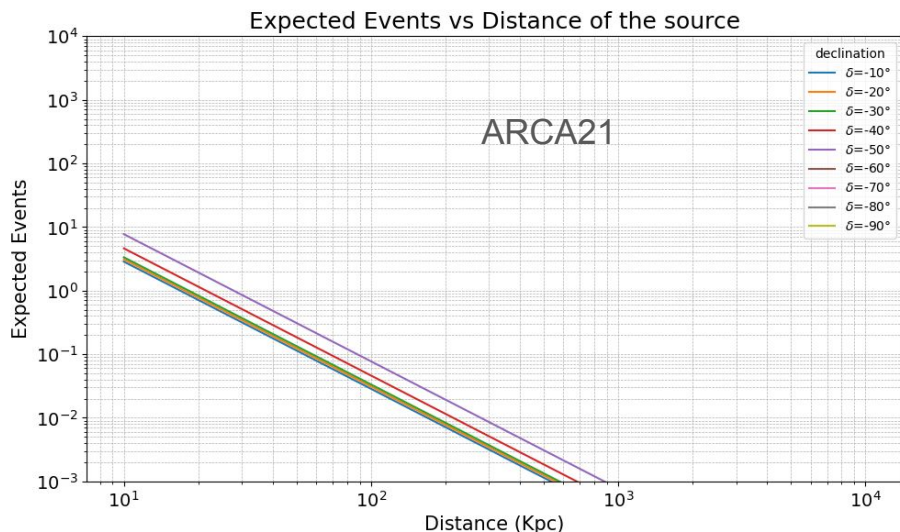
Number of expected events for sources at negative declination



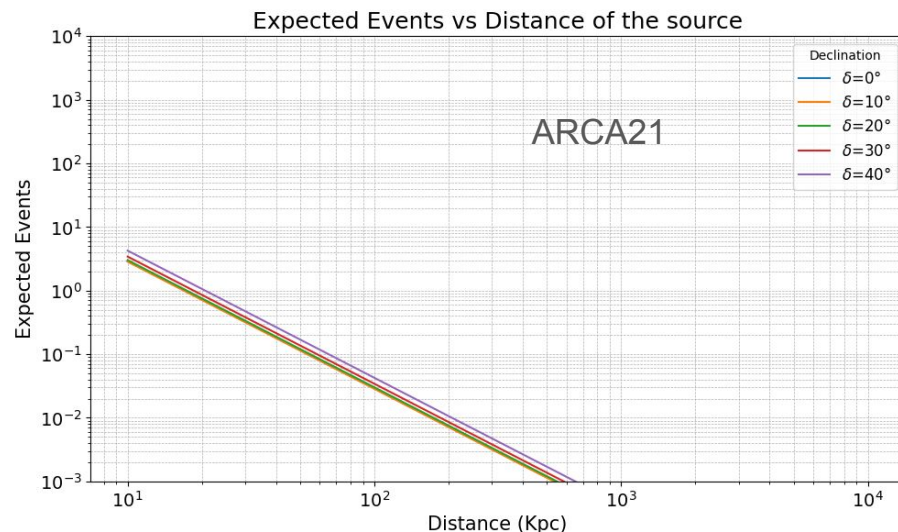
Number of expected events for sources at positive declination

2.b) Number of expected events (SN type IIP)

We chose to plot the number of events vs. the source distance in a range from 10 Kpc to 10 Mpc, for each declination → **SN type IIP**



Number of expected events for sources at negative declination



Number of expected events for sources at positive declination

2.c) Sources catalogue

At the moment, the catalogue contains a bunch of SNe exploded in the entire ARCA lifetime (from ARCA6 to ARCA21) but we need to select them more carefully according to each ARCA period!

SN_NAME	SN_TYPE	RA	DEC	Extended Source	Galactic SN	red_shift	distance (Mpc)	Alert data
SN2024 gyt II		140	0.9	0 No		0.018	76.0	20/4/2024 7:42
SN2024ggi II		170	-32.8	0 No		0.002435	10.3	11/4/2024 3:22
SN2024 gfc II		78	-68.4	0 No		0.012	50.7	4/4/2024 8:08
SN2024 fpb II		287	-17.9	0 No		0.022132	93.5	29/3/2024 11:19
SN2024 fpf II		215	22.0	0 No		0.0152	64.2	4/4/2024 12:40
SN2024 faq II		236	40.4	0 No		0.026	109.8	28/3/2024 10:40
SN2024exw II		183	14.8	0 No		0.007	29.6	24/3/2024 16:11
SN2024evl II		171	0.6	0 No		0.026364	111.3	22/3/2024 6:02
SN2024elf II		264	40.0	0 No		0.03	126.7	15/3/2024 13:42
SN2024ehs II		163	17.6	0 No		0.003776	15.9	15/3/2024 11:16
SN2024ees II		188	14.5	0 No		0.0235	99.2	11/3/2024 6:44
SN2024eek II		279	-62.5	0 No		0.015	63.3	11/3/2024 8:26
SN2024ecu II		284	-42.4	0 No		0.02	84.4	9/3/2024 0:45
SN2024ecr II		205	-16.6	0 No		0.052	219.6	10/3/2024 0:13
SN2024dym II		154	-27.4	0 No		0.014	59.1	5/3/2024 22:14
SN2021qqu II		54	-15.0	0 No		0.002	8.4	20/6/2021 10:19
SN2019fcn II		122	-28.1	0 No		0.003	12.7	8/5/2019 23:02

Only a few sources are closer than 10 Mpc!

2.d) Expected events from sources in the catalogue

Considering the closest sources from the catalogue, we tried to compute the number of expected events.

For the moment, we have selected only the SN whose H-E neutrino emission occurs in a time window compatible with ARCA21 life time.

	Declination (°)	Distance (Mpc)	#expected_events
SN2023ixf	54°	6.4 Mpc	10^{-7}



We will compute the number of expected events also for the SNe occurred during the life time of the other ARCA configurations (ARCA6, ARCA9 and ARCA19)

Next steps

- ❖ Upgrade the catalogue;
- ❖ Study of the sensitivity using Aaet unbinned likelihood;
- ❖ Perform a Cut&Count analysis.

Thanks for your attention!

Any suggestions?

Back up

• SNe HE neutrino energy fluence

The neutrino energy The neutrino flux The effective pp optical depth set to 1 for SNIIP and SN2n (depend on the cross section and proton inelasticity) The energy fraction carried by cosmic rays = 0.1 The proton momentum index (2.0 - 2.2)

$$E_\nu^2 \phi_\nu \sim \frac{1}{4\pi d^2} \frac{1}{2} \overbrace{\min[1, f_{pp}]} \frac{\epsilon_{cr} \epsilon_s}{R_{cr10}} \left(\frac{E_\nu}{0.4} \right)^{2-s}$$

The flux decreases with d over the surface area of a sphere

$R_{cr10} \equiv \epsilon_{cr} \epsilon_s / (E_p^2 dN_{cr}/dE_p)|_{10\text{GeV}}$

The proton energy The number of cosmic rays

The bolometric cosmic ray energy is divided by the differential cosmic ray energy at 10 GeV

$\epsilon_s \approx L_s t_{dyn}$

The kinetic luminosity The dynamical time

$L_s = 2\pi \rho_{cs} V^3 R_s^2$

The density profile of the CSM The velocity of the shell The shock radius

$t_{dyn} \approx \frac{R_s}{V_s}$

$V_s = \frac{dR_s}{dt}$

Back up

• Interaction of the supernova ejecta and a dense circumstellar material (CSM) (TeV-PeV)

Neutrino emission times between **2 hours** **1 year** post-core bounce.

$$pp \rightarrow \pi^+ \rightarrow \mu^+ \nu_\mu \rightarrow e^+ \nu_e \nu_\mu \bar{\nu}_\mu$$

CSM is characterised by its' **density profile**:

$$\rho_{cs} = Dr^{-w}$$

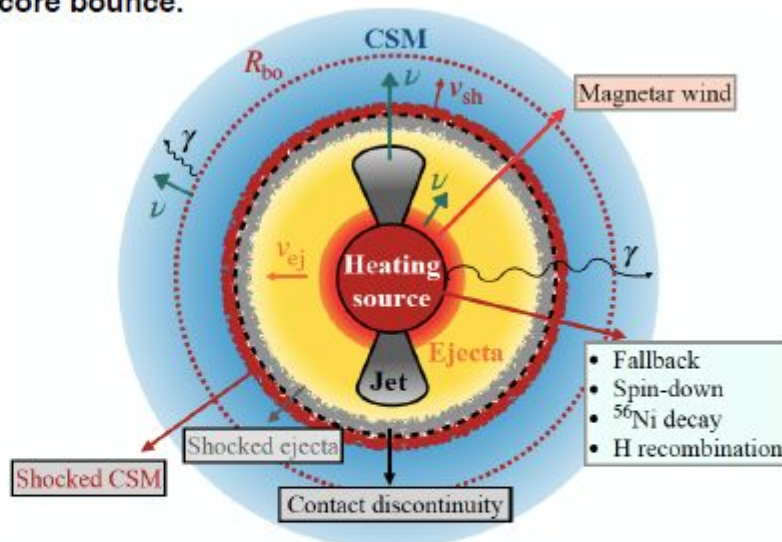
w wind profile ($w = 2$ reasonable in many cases)

In the wind case: $D = 5 \times 10^{16} \text{ g cm}^{-1} D_*$

$$D = \frac{\dot{M}_w}{4\pi V_w}, \text{ with } \dot{M}_w \text{ the mass loss rates and } V_w \text{ the wind velocity}$$

CSM parameters for various types of SNe

Class	D_*	$\dot{M}_w [M_\odot \text{ yr}^{-1}]$	$V_w [\text{km s}^{-1}]$	$R_* [\text{cm}]$
IIn	1	10^{-1}	100	10^{13}
II-P ^a	10^{-2}	10^{-3}	100	6×10^{13}
II-P ^b	1.34×10^{-4}	2×10^{-6}	15	6×10^{13}



Key parameters affecting the HE ν flux

- mass loss
- wind velocity
- shock velocity
- proton spectral index

Back up

- **Relativistic Choked jets (GeV-TeV)** jets of material trapped in the outer shells of the
Only high energy neutrinos are able to escape these jets
Neutrino emission times between 10 s to 10⁴ s.

In this scenario the proposed progenitors are similar to gamma-ray bursts (GRBs), except the jets are slower and never break through the stellar envelope, leading to neutrinos without the counterpart gamma-ray emission.

